The Ishango Bone as Recursion

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1 Introduction

This paper is a speculative archeological treatise. Herein lies an attempt at a proof that the Ishango bone is the earliest known example of recursion. The Ishango bone is at least 25,000 years old. Mathematical and calendrical theories are contenders that explain the purpose of the tally bone [2, 3]. The Ishango bone is speculated to be one of the earliest human artifacts of a mathematical or calendrical nature.

The theory that will be presented in this document, if true, would have far-reaching conclusions that early human thinking was capable of recursion. If false, then this theory is a projection of modern thinking onto an ancient artifact. A tally bone is a record keeping device but the Ishango bone is an artifact of a complex thought pattern. The difference between this theory and other theories is the difference between counting and recursion. Counting, primes, groupings, tables and recursion might suggest a mathematical knowledge but it is more likely a complex rule or set of rules that mimic mutual co-dependence.

A tally itself is a simplex because it relates a thing to an abstraction in a 1:1 relationship. In between two tallies is nothing and that is emphasized by the groupings in two orientations. The theme of groupings plays well into this document's argument. There is a gestalt here that needs to be examined.

The first justification is a modern take of a cultural classic which is the idea that the "medium is the message". Media theory supposes that human-made artifacts are extensions of the human body converted into different forms and functions. What is the form and function of this particular tally bone? Tallying to an extreme transforms mere counting into abstraction and the first abstraction are tally groupings because the groupings are themselves tallies.

The second justification is as a counter-factual. What if the tally bone is a tally of bones? Suppose that the Ishango bone is a tally of bones then the evidence must be gathered to support this claim. The counter-factual question grounds this theory. How is the human skeleton tallied?

The third justification is the tabular nature of the tallies [4]. This is the first clue which is our base case: there are three groups of tallies, and within a group are groups of tallies. Tabulation is a critical form in the Ishango bone, and separated groups of tallies suggest some ordering and counting.

Is this justified? There is no apparent contradiction so far. This and other perspectives allow for a leap from a simplex to a complex. There needs to be evidence and short of evidence we need a case.

The Ishango bone axial count is the first clue. There are two orientations of tally groupings on the Ishango bone: the first grouping is 60-48-60. This plays well into a base-12 number system. [4]

Let the human form be our reference. It is not suggested that the bone is human - the human form is only for reference. It is for comparison only.

The human skeleton is grouped by its forms and functions: head, torso, limbs. The natural question to ask is: how many bones are in each grouping? There are a total of 206 bones in the human body. The head has 29 bones. Each arm has 32 bones - only counting the humeris, forearm and hand then there are 30 bones. Each lower limb has 30 bones: 26 bones in the foot plus femur, patelis, tibia and fibula. The ribs (without sternum) and spine, have 24 bones each if you count the coccyx and sacrum as grouped with the pelvis (ileum, isceum and pubis). We would be amiss if we didn't tally 60 bones in an arm and leg, and 48 bones in ribs and spine. In the spirit of the Ishango theories, a mathematical presupposition of groupings of a skeletal form and function are being made. As this exercise progresses the case becomes overwhelming. If the tally bone is a tally of bones then it is the earliest example of recursion - a more complex extension of human reasoning.

The Ishango bone was found near Lake Edward in the Democratic Republic of the Congo and is generally dated to the Late Paleolithic, although estimates vary. Its groups of notches have prompted proposals involving tallying and numerical patterns, including prime numbers, but no interpretation is conclusive. [1] This paper therefore offers a speculative alternative: that the Ishango bone records an actual tally of a partial skeleton.

2 Transformations

Identify the anatomical groupings of the appendicular skeleton and the axial skeleton. There are 206 bones in the human body but the number of tallies is 168. Only a subset of bones are tallied on the Ishango bone. The subgroups are tallied and grouped by the bone count identified above as 60-48-60: limbs, ribs and spine.

The next step is to expand the tallies further: hands and feet do not yield tallies but it's the spine that has a possible connection to 7, 5 and 12. The spine tallies are key: those groups of tallies are grouped as representing

the spine. Ribs make up the rest of the tallies in mutually complementary groupings.

The middle column of tallies when grouped as ribs and spine share the space in an alternating pattern further broken down by mutual pairs of $n - 2n$ tallies.

However, the limb tallies do not follow $n - 2n$ mutual groupings because of the hand and foot tallies. Tallying each limb does not yield any Ishango tallies. Twenty-seven bones in the hand and 26 in the foot has no match on the Ishango bone. Twenty-one is the maximum number of tallies in a group.

The limb tallies are split between D and G columns. A further grouping does follow the $n - 4$ pattern. Two groupings that are mutually complementary where one pair of D tallies are equal to the other pair of G tallies. That leaves the other two pairs mutually complementary by $n - 4$ progression.

What has been identified so far?

Complementary pairings according to rules of arithmetic and groupings using skeletal structure as guide. Overall, the number of columns, groups and tallies are consistent with numerical reasoning beyond only counting. The mutual complementarity and $n - 2n$ and $n - 4$ are simple arithmetical rules but they are not trivial. There is an impression that these rules are consistent and another layer of abstraction is required to support the idea of recursion. What do we need to support this idea?

A series of tables are needed to construct the argument. Tabulating the skeletal structure according to the tallies is a first step. The Ishango bone is not a full representation of the human skeleton but it is a partial representation. The closest match is the appendicular skeleton and the axial skeleton with exceptions mentioned above. Placing the proposed skeletal structure into a table according to the G-M-D columns is a first step, Table 1. Proceeding to a true tally of the skeletal system is the next inferential step.

The M column matches on the spinal tally with the cervical and thoracic tallies leaving the lumbar tallies as the first complement $n - 2n$. The column M implies a complementary and mutually shared space: ribs-spine-ribs-spine. This is the pattern that is then followed with the limbs. Because the limbs are not distinguished they each have 30 tallies, the mutual complements are followed. Two pairs of tallies are identified as equal $11 - 19$ and $19 - 11$ which leaves the other pair as complementary $13 - 17$ and $9 - 21$. Overall, $n - 2$ holds for sequences 9, 11, 13 as well as 17, 19 and 21.

In the tabular progression, the limbs are identified as being mutual. The tallies are indicated.

It is observed that in the tabular tallies that each skeletal group is divided in a complementary fashion.

What happens when we swap rows to regroup the overall skeletal groupings?

Invert Table 4 so that the cervical tally is at the top, Table 5. Then swap the two middle rows to group spine with spine, and ribs with ribs, Table 6.

The inversion with middle row swap is the same as swapping first and last rows only. The middle group M of tallies has been regrouped - ribs are with ribs, and spine with spine.

What is obtained?

To see how the limbs are regrouped simplify the table for the limbs only by removing the middle column M. The simplified table shows the limbs in complementary tallies. Finally, stack the table to reveal an alternating pattern where each tally group of 30 are limbs. The patterns reveals pairs of co-dependent tallies.

3 Ishango Groupings on a K_{21} Graph

Given the tabular groupings of tallies, it was a natural step to a line graph. The first attempt at a graph was a table with vertical labels 1-21 and columns G-M-D. Then drawing lines from cell-to-cell revealed co-dependence.

Immediately following a horizontal-vertical orientation, a circular representation with 21 points along the circumference was drawn. Then the Ishango tallies were plotted.

The result were tally chords on a K_{21} graph Figure 2. Chords are drawn between points according to Tables 5 and 6. The chords are grouped according to sum modulo 21. There are three fix points: 6, 15, and 18.

Fix point 6, sum 12 (mod 21): tallies are 5-7, 4-8.

Fix point 15, sum 9 (mod 21): tallies are 3-6, 9-21, 11-19 (19-11) and 13-17.

Fix point 18, sum 15 (mod 21): tallies are 5-10.

As a reminder, the Ishango tallies are

$$3, 4, 5, 5, 6, 7, 8, 9, 10, 11, 11, 13, 17, 19, 19, 21.$$

The sum of the tallies is

$$2 \times (5 \times 9 + 2 \times 12 + 1 \times 15)$$

and also

$$4 \times 30 + 9 + 2 \times 12 + 1 \times 15,$$

which both equal 168. Moreover,

$$168 (\text{mod } 21) = 0$$

and

$$8 \times 21 = 168.$$

Furthermore, all possible chords on the K_{21} graph modulo 21 gives all the possible groupings of tallies.

How many possible chords of 8 groupings are there on a K_{21} graph that add to 168?

4 Tables and Figures

Region	G	M	D
	Upper Limb (30)	Ribs (24)	Upper Limb (Arm + Hand)
	Lower Limb (30)	Spine (24)	Lower Limb (Leg + Foot)
Count	60	48	60

Table 1: Minimal Groupings

Region	G	M	D
Top	Hand (27) Radius (1) Ulna (1) Humerus (1)	Ribs (24)	Hand Radius Ulna Humerus
Bottom	Femur (1) Patella (1) Tibia (1) Fibula (1) Foot (26)	Thoracic (12) Lumbar (5) Cervical (7)	Femur Patella Tibia Fibula Foot
Count	60	48	60

Table 2: Expanded Groupings

Region	G [Right]	M	D [Left]
Top	Hand (27) Radius (1) Ulna (1) Humerus (1)	Ribs (24)	Hand Radius Ulna Humerus
Bottom	Femur (1) Patella (1) Tibia (1) Fibula (1) Foot (26)	Thoracic (12) Lumbar (5) Cervical (7)	Femur Patella Tibia Fibula Foot
Count	60	48	60

Table 3: Re-orientation

Region	R	M	L
T1M1T1	R Limb G (11)	Ribs M1 (3+6)	L Limb G (11)
B2M0B4	R Limb D (13)	Thoracic (4+8)	L Limb D (21)
B3M1T2	R Limb D (17)	Ribs M1 (10+5)	L Limb G (19)
T2M0B1	R Limb G (19)	Lumbar (5) Cervical (7)	L Limb D (9)
Count	60	48	60

Table 4: Ishango Bone Tabulation

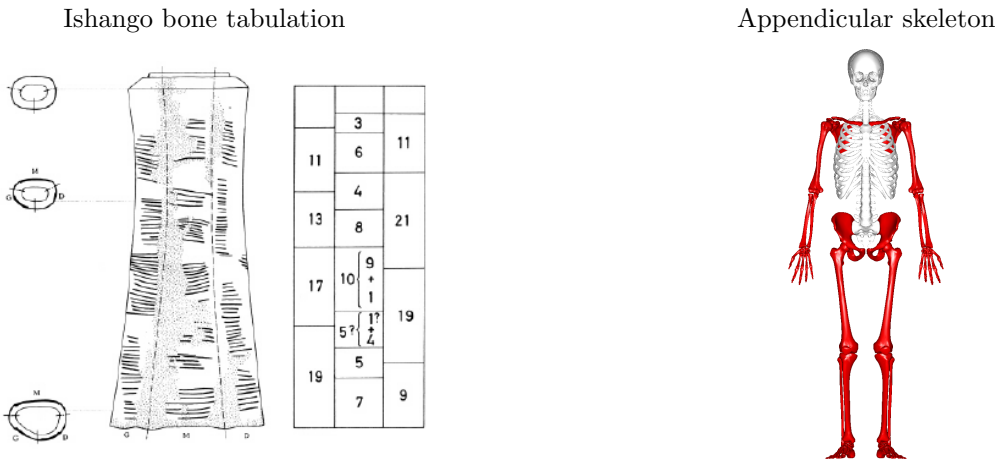


Figure 1: Comparison between the Ishango bone [4] and the appendicular skeleton [5].

Count	60	48	60
T2M0B1	R Limb G (19)	Cervical (7) Lumbar (5)	L Limb D (9)
B3M1T2	R Limb D (17)	Ribs M2 (5+10)	L Limb G (19)
B2M0B4	R Limb D (13)	Thoracic (8+4)	L Limb D (21)
T1M1T1	R Limb G (11)	Ribs M1 (6+3)	L Limb G (11)
Region	R	M	L

Table 5: Inverted Ishango Bone Tabulation

Count	60	48	60
T2M0B1	R Limb G (19)	Cervical (7) Lumbar (5)	L Limb D (9)
B2M0B4	R Limb D (13)	Thoracic (8+4)	L Limb D (21)
B3M1T2	R Limb D (17)	Ribs M2 (5+10)	L Limb G (19)
T1M1T1	R Limb G (11)	Ribs M1 (6+3)	L Limb G (11)
Region	R	M	L

Table 6: Middle Rows Swapping

G (19)	D (9)
D (13)	D (21)
D (17)	G (19)
G (11)	G (11)

Table 7: Limb Tallies

G (19)	D (9)
D (13)	D (21)
D (17)	G (19)
G (11)	G (11)
G (19)	D (9)
D (13)	D (21)
D (17)	G (19)
G (11)	G (11)
G (19)	D (9)
D (13)	D (21)
D (17)	G (19)
G (11)	G (11)

Table 8: Repeated Tabulation

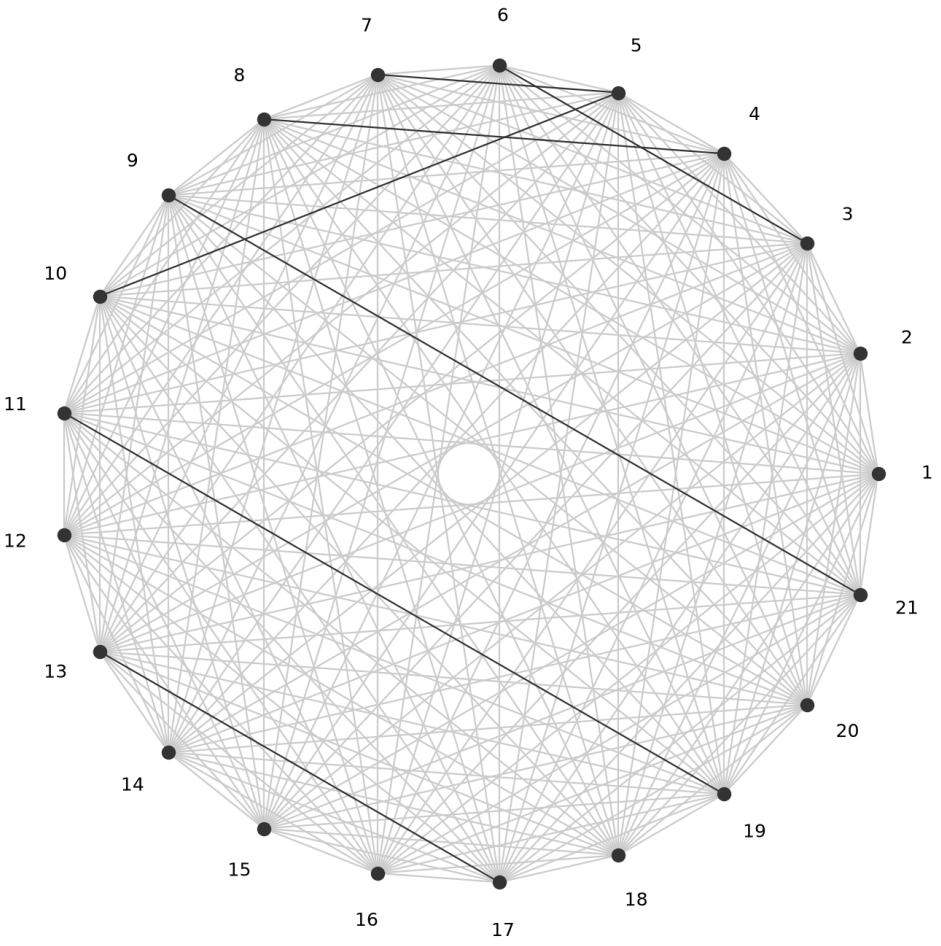


Figure 2: Tally groupings on a K_{21} graph

5 Conclusion

The distribution of tallies follows a pattern of paired co-dependence of groupings. Using the theory of recursion, isolating the appendicular and axial skeleton as the distribution of numbers on the Ishango bone reveals a numerical and logical consistency. Using modern principles and anatomical knowledge, the Ishango bone becomes a telling mathematical artifact. From a speculative perspective into the archeological artefact of the Ishango bone, it is a natural progression from counting to recursion.

The final conclusion is that the Ishango bone is possibly the oldest known example of recursion by early humans and features a partial skeletal anatomy in gait.

References

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